FL-06

Prompting Fundamentals on Real Tasks

FlyRank AI Fluency Program – Week 2

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Submission: FL-06 Prompting Fundamentals on Real Tasks

Assignment Overview

This assignment focused on improving prompt quality through iterative refinement, using prompt engineering techniques applied to a real-world documentation task. The exercise involved writing and progressively revising a single prompt — generating a recruiter-friendly GitHub README — to observe how specific prompting techniques change the accuracy, structure, and usability of AI-generated output.

Objectives

- Apply prompt engineering techniques to a practical documentation task
- Compare prompt quality and output quality across successive iterations
- Compare Claude and ChatGPT responses to the same task
- Create a reusable prompt template for future projects
- Document improvements through structured experimentation

Real Task

Write a recruiter-friendly GitHub README for an AI Internship Landing Page project.

Prompt Iteration Process

Baseline Prompt

Prompt

```
Write a README for my GitHub project.
```

Output Summary: The model could not generate a README and instead asked for missing project details (name, description, tech stack, features, setup steps).

What Changed: No persona, no context, and no project information were provided.

What Improved: N/A — this was the starting point for comparison.

What Still Failed: No usable draft was produced; the request was too underspecified to act on.

Next Improvement: Introduce a persona to see whether framing the assistant's role changes response quality.

Version 1 – Role Assignment

Prompt

```
You are an experienced technical writer.
Write a README for my GitHub project.
```

Output Summary: The model still asked clarifying questions before drafting, but the tone of the questions became more consultative and structured.

What Changed: Added an explicit persona ("experienced technical writer").

What Improved: Slightly more professional framing of the follow-up questions.

What Still Failed: Persona alone did not supply the missing project facts, so no draft was generated.

Next Improvement: Add context about the project's purpose and audience to reduce ambiguity.

Version 2 – Context and Motivation

Prompt

```
You are an experienced technical writer.
This README is for my AI Internship Landing Page project. It will be part
of my GitHub portfolio and will be reviewed by recruiters for Frontend
Developer and Software Development Internship roles.
Write a README for this project.
```

Output Summary: The model asked targeted clarifying questions specifically about tech stack and deployment status, using the recruiter context to prioritize what mattered most.

What Changed: Added project name, portfolio purpose, and target audience (recruiters).

What Improved: Follow-up questions became relevant and scoped instead of generic.

What Still Failed: Tech stack and content specifics were still missing, so a draft could not yet be produced.

Next Improvement: Provide an explicit structural template and concrete tech stack information.

Version 3 – Few-shot Examples

Prompt

```
This project is an AI Internship Landing Page built using HTML5, CSS3,
and JavaScript.
Here is an example of the structure I want:
Project Overview / Features / Tech Stack / Getting Started / Author / License
Now write a README for my project using the same structure.
```

Output Summary: The model produced a complete, usable README draft as a downloadable file, following the example heading structure, though some feature details were reasonably assumed.

What Changed: Supplied a concrete tech stack and an example section structure to imitate.

What Improved: First full draft generated; structure matched the requested example exactly.

What Still Failed: Some content (features, project structure) was generic/assumed rather than grounded in verified project details.

Next Improvement: Enforce an exact structure and add rules restricting the model from inventing details.

Version 4 – Output Structure

Prompt

```
Create a professional GitHub README.
Use exactly this structure:
# Project Name / ## Project Overview / ## Features / ## Tech Stack /
## Project Structure / ## Getting Started / ## Author / ## License
Only use the information provided. Do not invent technologies, features,
links, structure, or installation steps. Return the result only in Markdown.
```

Output Summary: Output matched the exact heading hierarchy requested, stayed within Markdown formatting, and marked unsupplied details (links, structure, license) as "Not provided" instead of inventing them.

What Changed: Locked the exact heading structure, added strict anti-fabrication rules, and constrained the output format.

What Improved: Draft became fully aligned to the required structure with no invented facts.

What Still Failed: Some sections (e.g., Getting Started, Project Structure) were left sparse since real data still hadn't been supplied — a data limitation, not a prompting one.

Next Improvement: Add explicit step-by-step reasoning to make the drafting process more transparent and deliberate.

Version 5 – Step Decomposition

Prompt

```
Work step by step:
Step 1: Understand the project from the information provided.
Step 2: Identify the important sections a recruiter expects.
Step 3: Draft each section one by one.
Step 4: Review the README for clarity, consistency, and completeness.
Step 5: Output the final README in Markdown.
```

Output Summary: The model explicitly worked through understanding the project, identifying recruiter-relevant sections, drafting, and reviewing before producing the final README — resulting in the most accurate and complete version.

What Changed: Broke the single instruction into five explicit reasoning steps performed in sequence.

What Improved: Final output was well-organized, fully grounded in supplied facts, and required no further correction.

What Still Failed: Optional fields (GitHub/LinkedIn links, project structure) remained "Not provided" since that data was genuinely never supplied.

Next Improvement: Combine step decomposition with a reusable template to standardize this workflow for future documentation tasks.

Cross-Model Comparison

The table below compares Claude and ChatGPT on the same README-writing task, based on general, commonly observed differences in how each model tends to handle structured writing requests.

Criteria	Claude	ChatGPT
Tone	Professional, measured, tends to ask clarifying questions before assuming details	Confident, fluent, often proceeds directly with plausible assumptions
Structure	Follows requested heading structure closely and consistently across edits	Generally follows structure, but may reorder or merge sections
Accuracy	More conservative about inventing unsupplied details when instructed not to	May fill gaps with plausible-sounding but unverified details unless explicitly restricted
Strengths	Strong instruction-following on formatting and constraints; good at flagging missing information	Fast, fluent prose; good at producing a complete-feeling draft quickly
Limitations	May ask more clarifying questions before producing a draft	May require more explicit correction to remove invented details

Note: This comparison reflects general tendencies observed across typical use, not a controlled side-by-side test of identical outputs.

Final Reusable Prompt

The final version below consolidates every improvement from the iteration process into a single reusable template for future README-writing tasks.

```
You are an experienced technical writer.
Create a professional GitHub README for the following project.
Follow exactly this structure:
# Project Name
## Project Overview
## Features
## Tech Stack
## Project Structure
## Getting Started
## Author
```

```
## License

Requirements:
- Write in professional Markdown.
- Keep the language concise and recruiter-friendly.
- Only use the information I provide.
- Do not invent technologies, features, links, repository structure,
  installation steps, or deployment details.
- If information is missing, clearly state "Not provided."
- Make the README suitable for a public GitHub repository.

Project Details:
Project Name: [Insert project name]
Project Overview: [1-3 sentences on what the project is and how it was built]
Features: [List key features]
Tech Stack: [List technologies used]
Project Structure: [Folder/file layout, or "Not provided."]
Getting Started: [Setup/installation steps, or "Not provided."]
Author: [Name, GitHub link, LinkedIn link, or "Not provided."]
License: [License type, or "Not provided."]
```

Learning Outcomes

- Learned iterative prompt engineering by refining one prompt across multiple versions
- Understood the effect of changing one variable (persona, context, structure, or reasoning steps) at a time
- Compared outputs across different AI models to identify strengths and limitations
- Created reusable prompt templates that generalize beyond a single project
- Improved an AI-assisted documentation workflow suitable for portfolio-quality output

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Conclusion

Structured prompting made a measurable difference in the quality and consistency of AI-generated documentation. Moving from a vague, single-line request to a prompt with clear persona, context, structure, explicit constraints, and step-by-step reasoning transformed the output from unusable clarifying questions into a precise, recruiter-ready README. The exercise reinforced that small, deliberate changes to a prompt — rather than simply asking for "more" — are what reliably improve AI-assisted work.